Task 1: Simulation and Sensitivity Analysis of Chaotic Systems

Ibrahim Hanafy

July 2026

1 Part A — Discrete Chaotic Systems

1.1 Logistic Map

The logistic map $x_{n+1} = r \cdot x_n(1 - x_n)$ was iterated for 1000 steps with $x_0 = 0.5$. At $r = 3.9$ (chaotic regime), the time series exhibits aperiodic oscillations bounded within $[0, 1]$, characteristic of deterministic chaos.

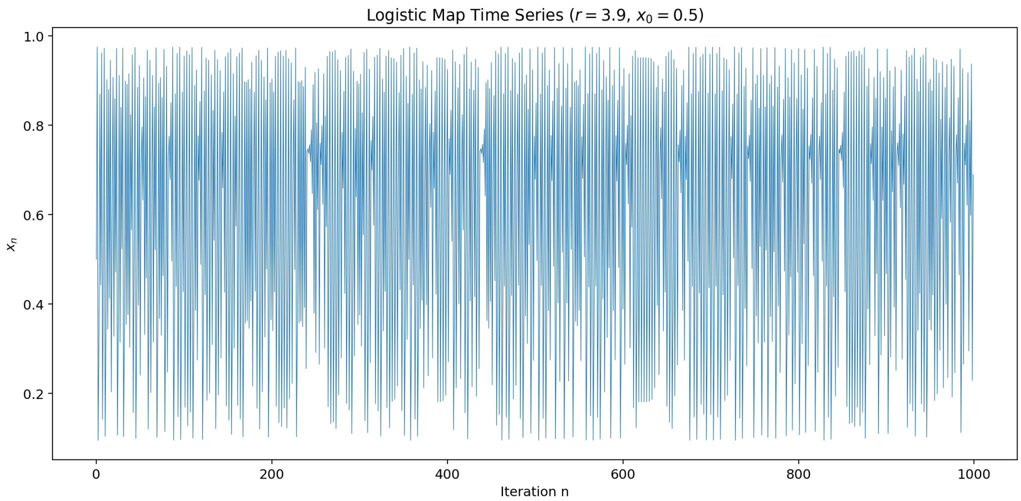

Figure 1: Logistic map time series at $r = 3.9$.

The bifurcation diagram (Fig. 2) sweeps $r \in [2.5, 4.0]$ with 2000 values, discarding the first 500 transient iterations. A clear period-doubling cascade is visible: the system transitions from a stable fixed point ($r < 3$) through successive period-doublings ($r \approx 3.0, 3.45, 3.54$) into full chaos ($r \gtrsim 3.57$). Periodic windows (e.g., the period-3 window near $r \approx 3.83$) appear embedded within the chaotic bands.

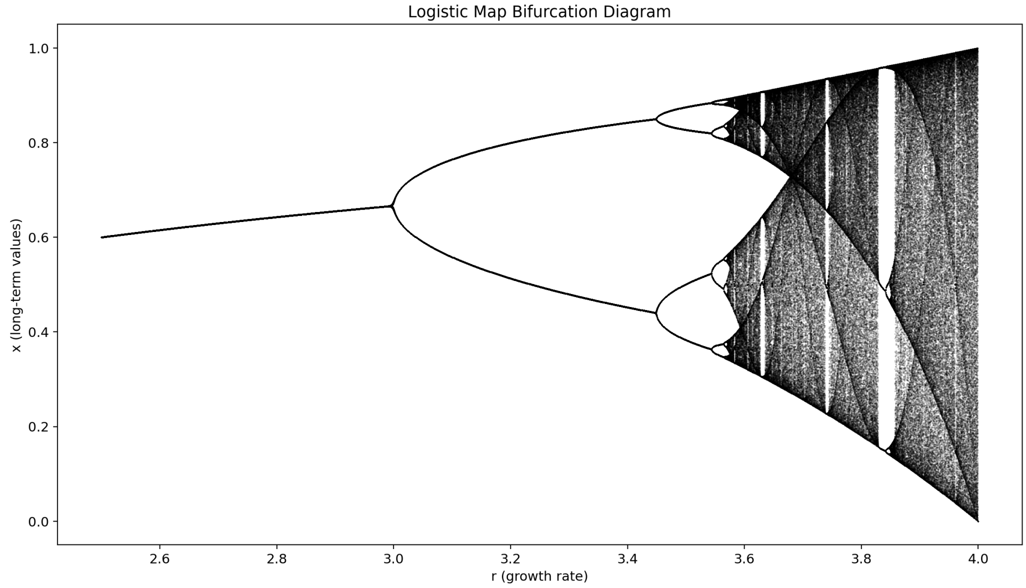

Figure 2: Logistic map bifurcation diagram.

1.2 Hénon Map

The Hénon map ($x_{n+1} = 1 - ax_n^2 + y_n$, $y_{n+1} = bx_n$) was iterated for 10000 steps at $a = 1.4$, $b = 0.3$ from $(x_0, y_0) = (0, 0)$. After discarding 500 transients, the attractor reveals the characteristic fractal banana-shaped structure with fine parallel striations, confirming the folding-and-stretching mechanism underlying the map's chaotic dynamics.

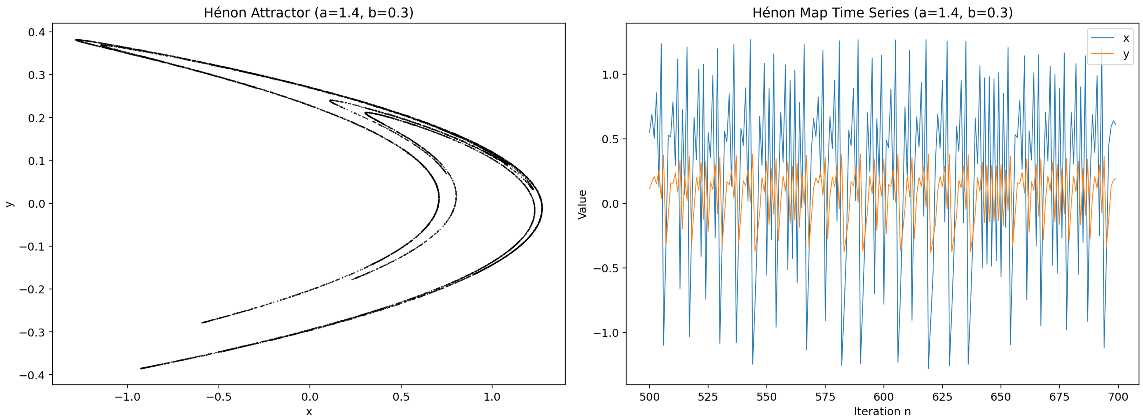

Figure 3: Hénon attractor (left) and time series (right) at $a = 1.4$, $b = 0.3$.

The bifurcation diagram (Fig. 4) sweeps $a \in [1.0, 1.4]$ with $b = 0.3$ fixed. Period-doubling routes to chaos are visible, similar in structure to the logistic map but with additional complexity from the two-dimensional dynamics. Some parameter values cause orbits to diverge to infinity.

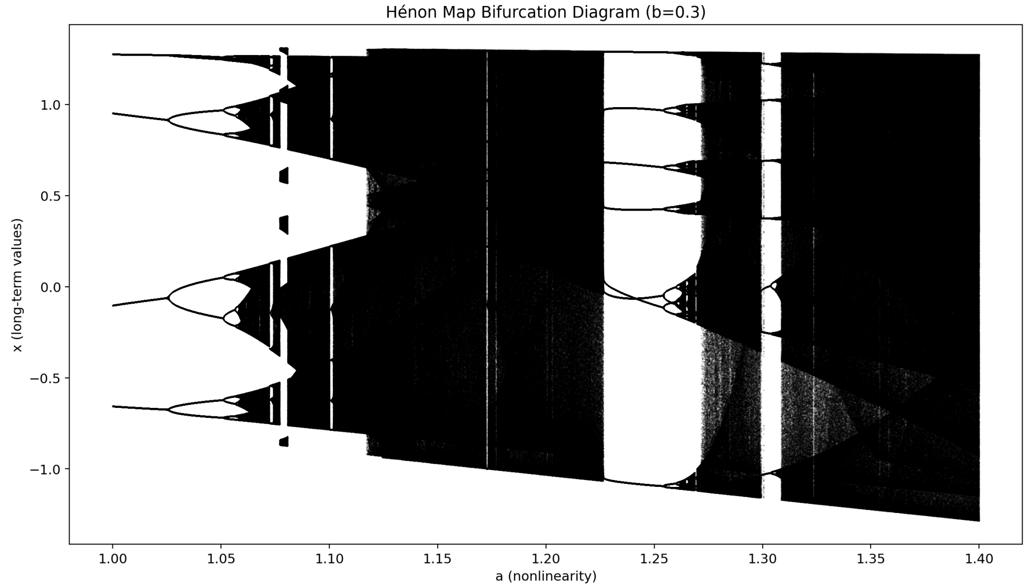

Figure 4: Hénon map bifurcation diagram ($b = 0.3$).

2 Part B — Continuous Chaotic Systems

All continuous systems were integrated using `scipy.integrate.solve_ivp` with the RK45 adaptive method (`rtol`=10^{-10}, `atol`=10^{-12}).

2.1 Lorenz System

Table 1: Lorenz system settings.

Setting	**Value**
Equations	$\dot{x} = \sigma(y - x)$, $\dot{y} = x(\rho - z) - y$, $\dot{z} = xy - \beta z$
Parameters	$\sigma = 10$, $\rho = 28$, $\beta = 8/3$
Initial condition	$(1, 1, 1)$
dt / T	0.01 / 50

The 3D phase-space trajectory (Fig. 5) forms the well-known two-winged butterfly attractor. The trajectory spirals around one lobe before switching unpredictably to the other. Time series (Fig. 6) confirm the irregular switching in $x(t)$ and $y(t)$, while $z(t)$ remains strictly positive.

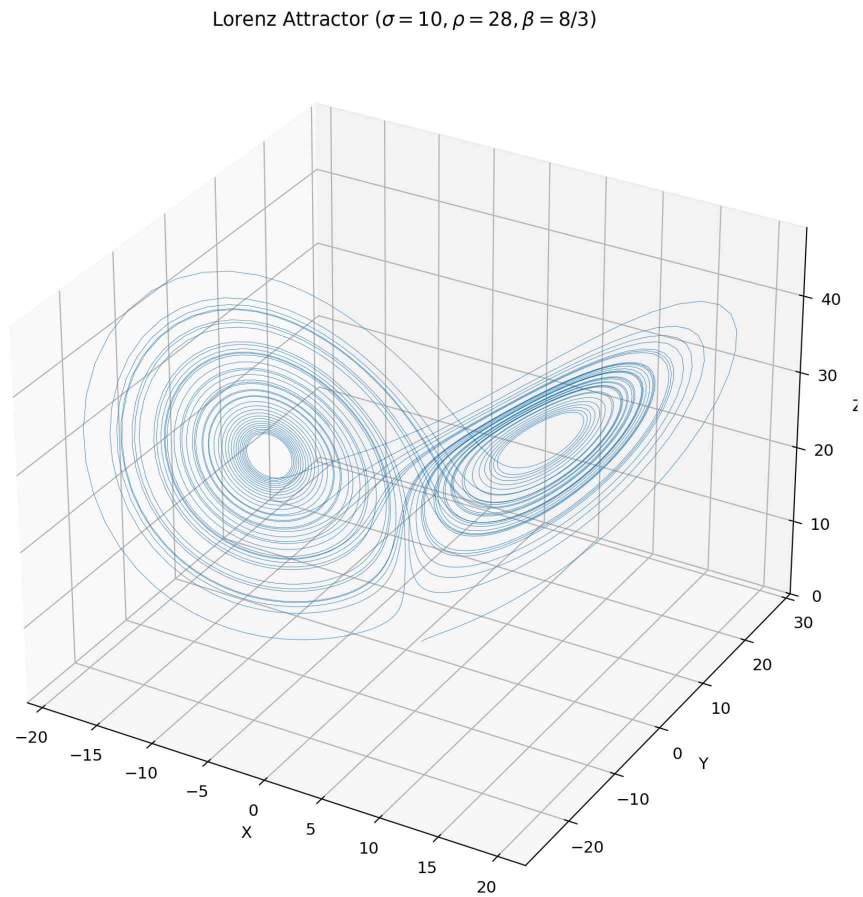

Figure 5: Lorenz attractor in 3D phase space.

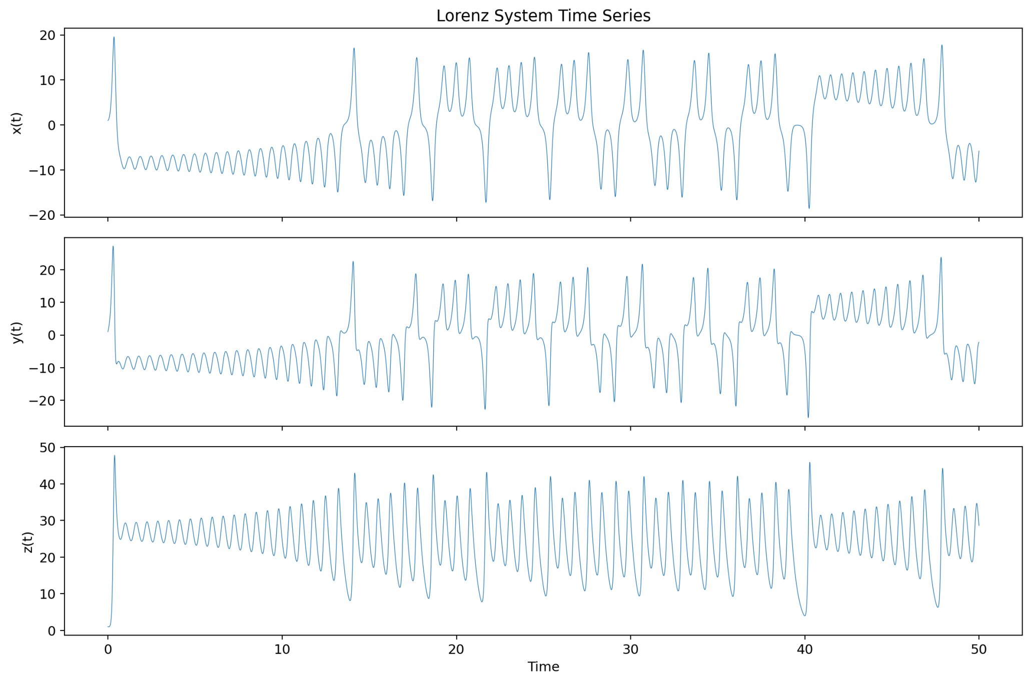

Figure 6: Lorenz system time series.

2.2 Rössler System

Table 2: Rössler system settings.

Setting	Value
Equations	$\dot{x} = -y - z$, $\dot{y} = x + ay$, $\dot{z} = b + z(x - c)$
Parameters	$a = 0.2$, $b = 0.2$, $c = 5.7$
Initial condition	$(1, 1, 1)$
dt / T	0.01 / 300

The Rössler attractor (Fig. 7) exhibits a single-scroll spiral in the x-y plane, with occasional upward excursions in z (the reinjection mechanism). Unlike the Lorenz system's double-scroll structure, Rössler produces a simpler topology with one positive Lyapunov exponent.

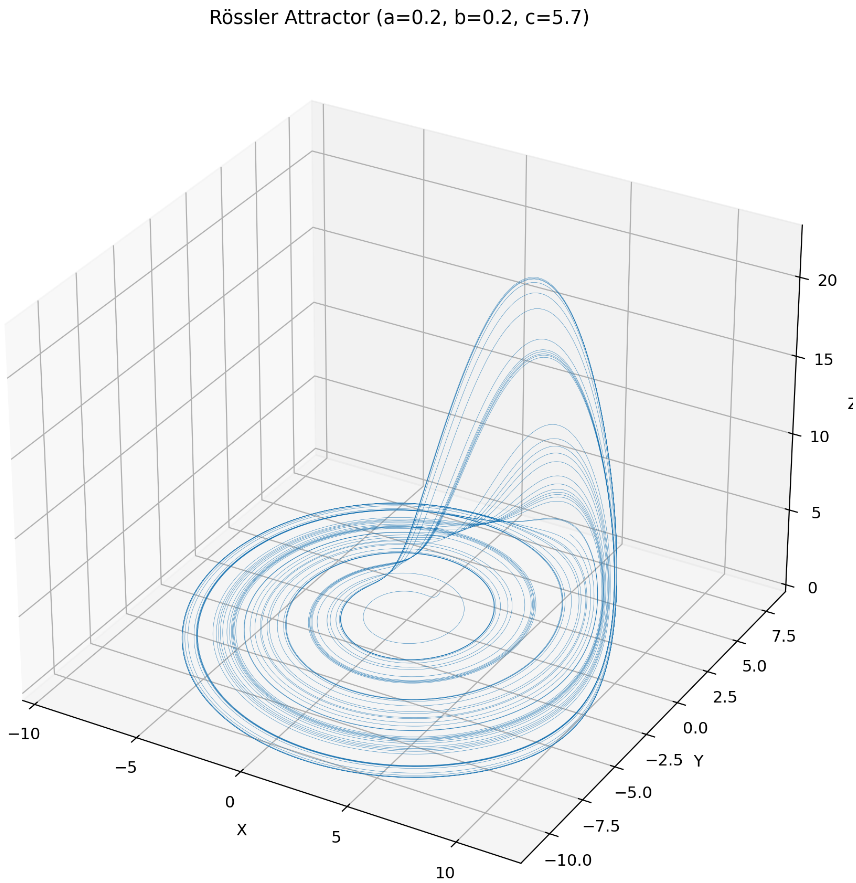

Figure 7: Rössler attractor.

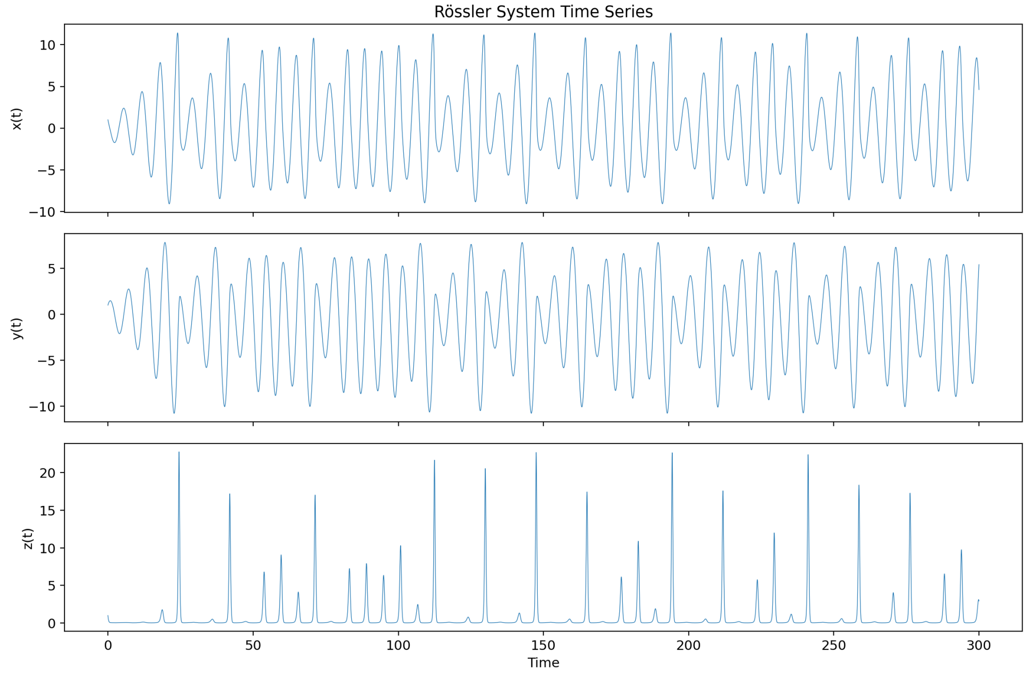

Figure 8: Rössler system time series.

2.3 Chen System

Table 3: Chen system settings.

Setting	**Value**
Equations	$\dot{x} = a(y - x)$, $\dot{y} = (c - a)x - xz + cy$, $\dot{z} = xy - bz$
Parameters	$a = 35$, $b = 3$, $c = 28$
Initial condition	$(-0.1, 0.5, -0.6)$
dt / T	0.002 / 50

The Chen attractor (Fig. 9) produces a double-scroll structure reminiscent of the Lorenz attractor but wider and more symmetric. Despite sharing the same $\dot{x} = a(y - x)$ form, the different coupling in $\dot{y}$ and $\dot{z}$ produces a distinct attractor geometry.

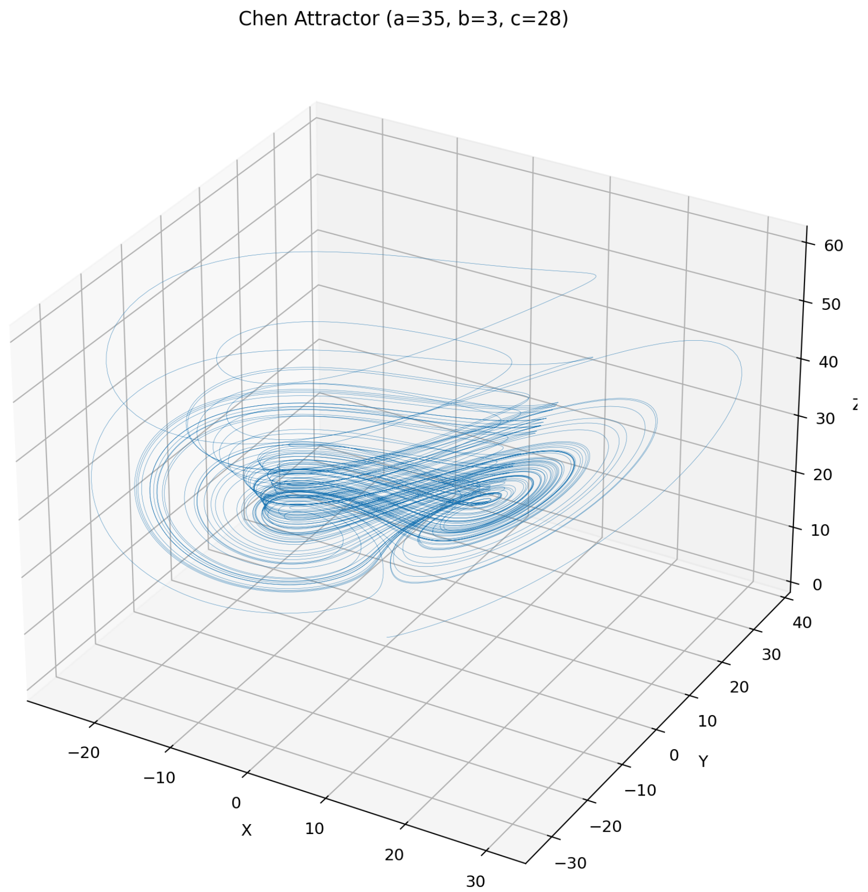

Figure 9: Chen attractor.

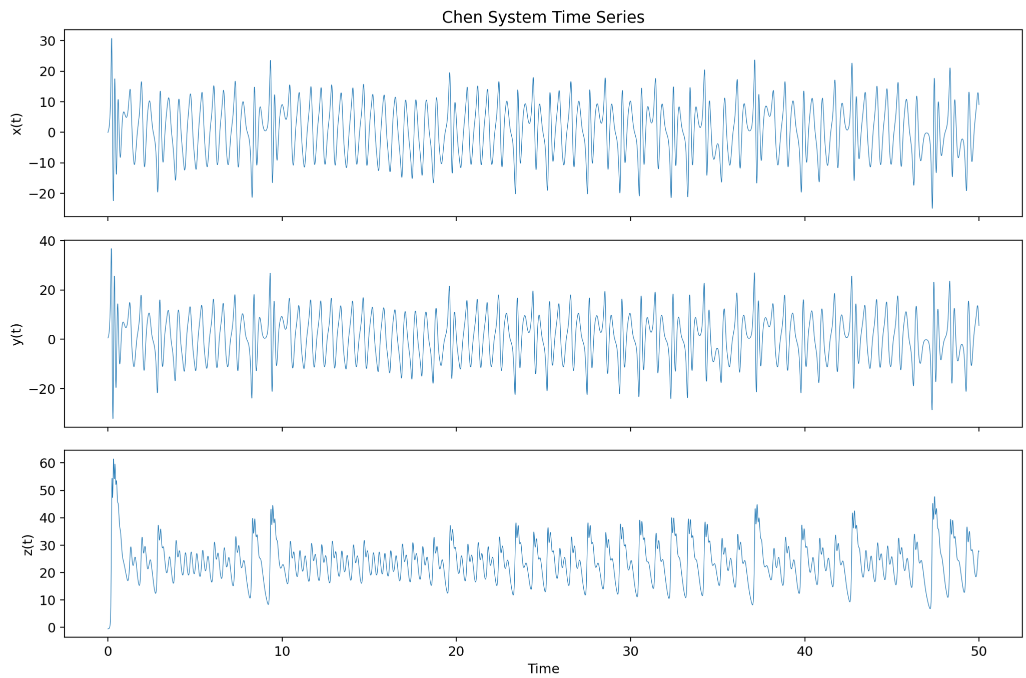

Figure 10: Chen system time series.

2.4 Hyperchaotic Rössler System

Table 4: Hyperchaotic Rössler system settings.

Setting	**Value**
Equations	$\dot{x} = -y - z,\ \dot{y} = x + ay + w,\ \dot{z} = b + xz,\ \dot{w} = -cz + dw$
Parameters	$a = 0.25,\ b = 3,\ c = 0.5,\ d = 0.05$
Initial condition	$(-10, -6, 0, 10)$
dt / T	0.01 / 250

The 4D hyperchaotic Rössler system requires phase-space projections for visualization. Figure 11 shows the x-y, x-z, and y-w projections. Compared to the standard Rössler attractor, the trajectories fill space more densely — a signature of hyperchaos, where two or more positive Lyapunov exponents cause divergence in multiple independent directions simultaneously.

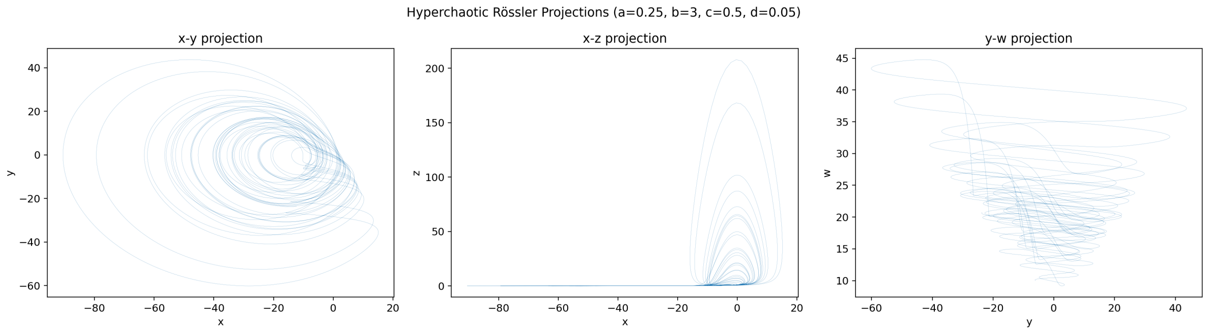

Figure 11: Hyperchaotic Rössler phase-space projections.

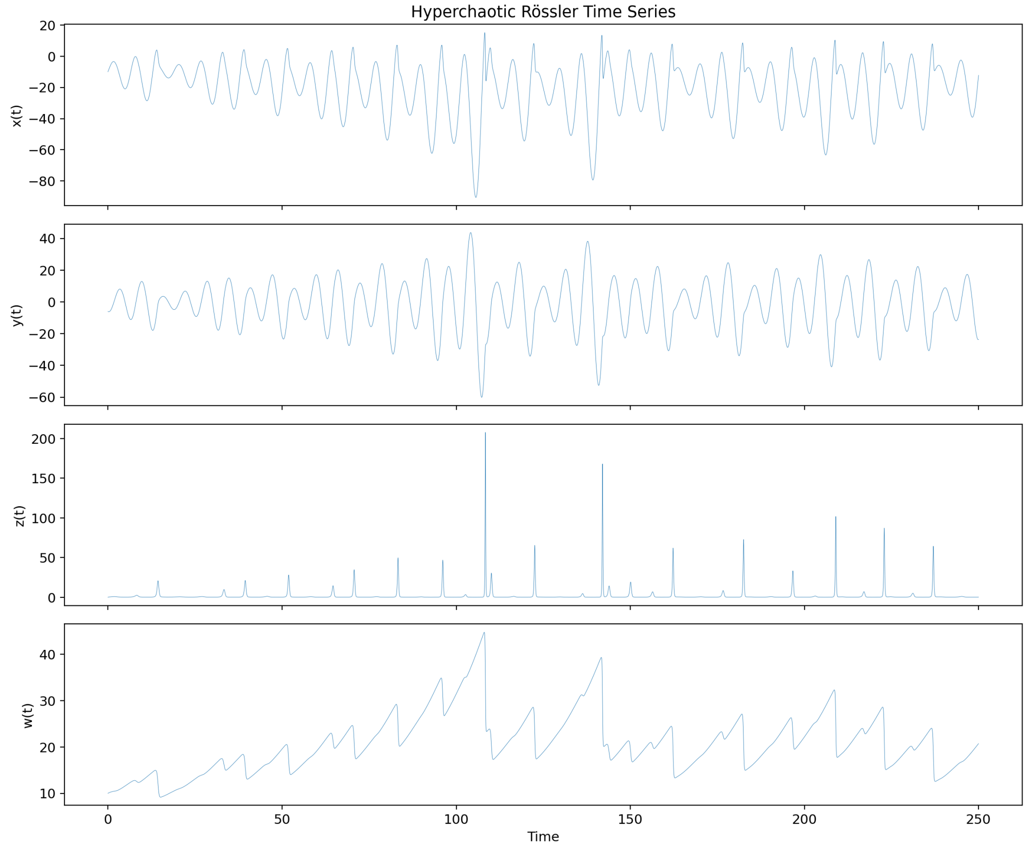

Figure 12: Hyperchaotic Rössler time series (all four state variables).

3 Part C — Sensitivity Analysis

3.1 Parameter Sweep: Lorenz ρ-Sweep

Figure 13 shows the local maxima of $z(t)$ as ρ is swept from 0 to 30 (with $\sigma = 10$, $\beta = 8/3$ fixed). For small ρ, the system converges to a stable fixed point (origin). Around $\rho \approx 1$, a pitchfork bifurcation creates two symmetric fixed points. Near $\rho \approx 24.7$, a Hopf bifurcation destabilizes these fixed points, and the system transitions through periodic windows into fully developed chaos at $\rho = 28$.

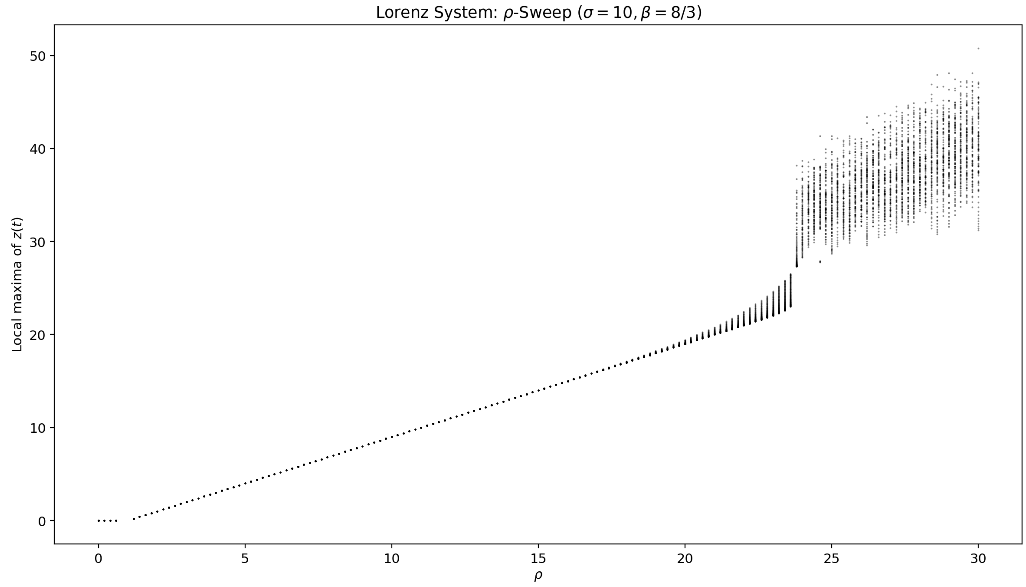

Figure 13: Lorenz ρ-sweep: local maxima of $z(t)$ vs. ρ.

3.2 Step-Size Sensitivity

The Lorenz system was re-simulated at the standard chaotic parameters using a **fixed-step RK4** integrator with $dt = 0.001$ (fine), $dt = 0.01$ (baseline), and $dt = 0.05$ (coarse). All three simulations start from the identical initial condition $(1, 1, 1)$.

As shown in Fig. 14, all three trajectories agree closely for the first ∼15–20 time units, after which they progressively diverge. The coarser step size ($dt = 0.05$) diverges earliest. The log-scale divergence plot (Fig. 16) reveals approximately exponential growth of the trajectory difference, consistent with the positive Lyapunov exponent of the system.

This demonstrates that in chaotic systems, numerical truncation errors at each integration step act as effective perturbations to the initial condition. Because nearby trajectories separate exponentially, even small differences in numerical accuracy lead to completely uncorrelated trajectories after a finite time. The $dt = 0.01$ step size is sufficient to resolve the attractor's geometry accurately, while $dt = 0.05$ begins to distort the attractor structure.

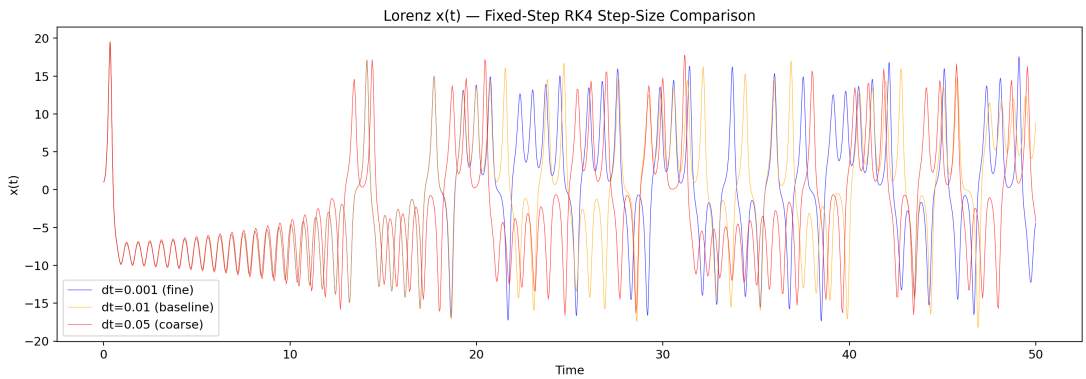

Figure 14: Lorenz $x(t)$ for three RK4 step sizes.

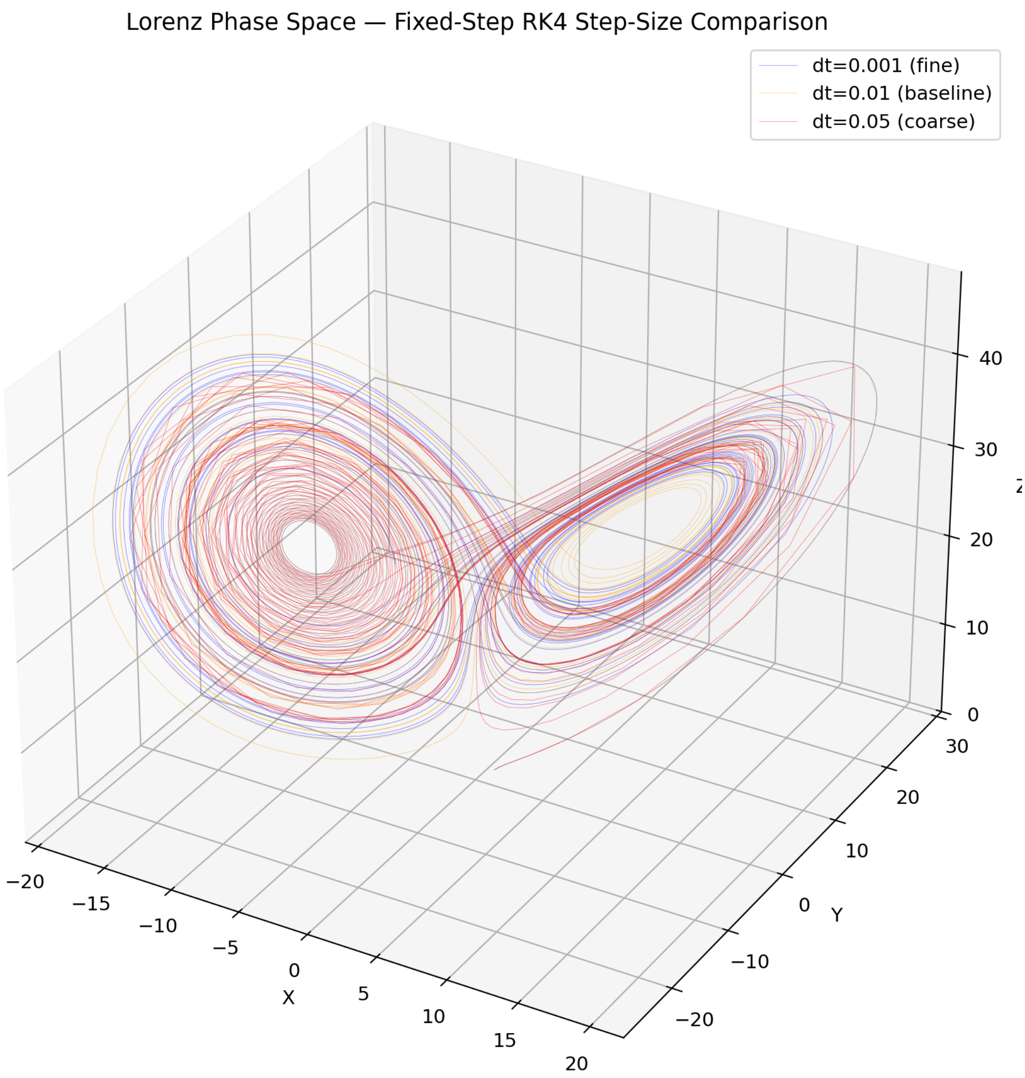

Figure 15: Lorenz 3D phase space overlay for three step sizes.

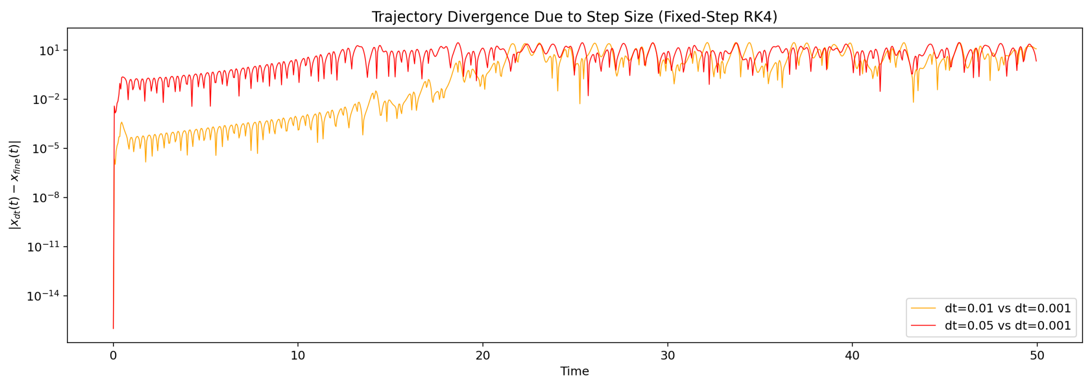

Figure 16: Trajectory divergence $|x_{dt}(t) - x_{\text{fine}}(t)|$ on a log scale.

4 Conclusion

Discrete chaotic systems (logistic and Hénon maps) demonstrate that chaos can emerge from simple iterative rules in as few as one dimension. Continuous chaotic systems (Lorenz, Rössler, Chen) require a minimum of three dimensions and exhibit richer dynamics including strange attractors with complex geometric structure. Both classes share the defining property of sensitive dependence on initial conditions, as confirmed by the step-size sensitivity analysis.

The transition from chaotic to hyperchaotic behavior, illustrated by the 4D Rössler system, involves the appearance of a second positive Lyapunov exponent. This causes trajectories to diverge in multiple independent directions simultaneously, producing denser, more space-filling attractors that are fundamentally harder to predict or reconstruct from partial observations.